

ROC Curve and AUC

CSE427: Machine Learning

Why Do We Need ROC?

Many classifiers output probabilities:

$$P(y = 1 \mid x)$$

To convert probability into a class, we choose a threshold:

$$P(y = 1 \mid x) \geq \text{threshold} \Rightarrow \text{Predict Positive}$$

$$P(y = 1 \mid x) < \text{threshold} \Rightarrow \text{Predict Negative}$$

Problem

Different thresholds produce different confusion matrices.

ROC Idea

ROC shows model performance across many thresholds.

Confusion Matrix Review

	Predicted Yes	Predicted No
Actual Yes	TP	FN
Actual No	FP	TN

TP = actual positive predicted positive

FN = actual positive predicted negative

FP = actual negative predicted positive

TN = actual negative predicted negative

Two Main Rates Used in ROC

ROC uses:

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

Metric	Meaning	Also Called
TPR	How many actual positives are found	Recall / Sensitivity
FPR	How many actual negatives are wrongly predicted positive	1-Specificity

ROC Point

Each threshold gives one point:

(FPR, TPR)

Example Data: 5 Positives and 4 Negatives

Suppose we have:

5 actual positives, 4 actual negatives

The classifier gives probabilities.

We change the threshold and calculate:

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

Goal

Observe how the ROC curve is created threshold by threshold.

Threshold = 1

At threshold 1, nobody is predicted positive.

	Predicted Yes	Predicted No
Actual Yes	0	5
Actual No	0	4

$$TPR = \frac{0}{0 + 5} = 0$$

$$FPR = \frac{0}{0 + 4} = 0$$

So the ROC point is:

$$(FPR, TPR) = (0, 0)$$

Threshold = 0.975

Now one actual positive is predicted positive.

	Predicted Yes	Predicted No
Actual Yes	1	4
Actual No	0	4

$$TPR = \frac{1}{1 + 4} = 0.2$$

$$FPR = \frac{0}{0 + 4} = 0$$

ROC point:

(0, 0.2)

Meaning

We found more positives without adding false positives.

Threshold = 0.965

Now two actual positives are predicted positive.

	Predicted Yes	Predicted No
Actual Yes	2	3
Actual No	0	4

$$TPR = \frac{2}{2 + 3} = 0.4$$

$$FPR = \frac{0}{0 + 4} = 0$$

ROC point:

(0, 0.4)

More Thresholds

As threshold decreases, more samples are predicted positive.

Threshold	TP	FN	FP	TN
1.000	0	5	0	4
0.975	1	4	0	4
0.965	2	3	0	4
0.950	3	2	0	4
0.870	3	2	1	3
0.000	5	0	4	0

For each row:

$$ROC\ point = (FPR, TPR)$$

ROC Points Table

Threshold	FPR	TPR
1.000	$0/4 = 0$	$0/5 = 0$
0.975	$0/4 = 0$	$1/5 = 0.2$
0.965	$0/4 = 0$	$2/5 = 0.4$
0.950	$0/4 = 0$	$3/5 = 0.6$
0.870	$1/4 = 0.25$	$3/5 = 0.6$
0.000	$4/4 = 1$	$5/5 = 1$

Plot these points:

$(0, 0)$, $(0, 0.2)$, $(0, 0.4)$, $(0, 0.6)$, $(0.25, 0.6)$, $(1, 1)$

What Does the ROC Curve Show?

ROC graph:

$x\text{-axis} = FPR$

$y\text{-axis} = TPR$

Good Classifier

High TPR with low FPR.

Bad Classifier

TPR increases only when FPR also increases a lot.

The diagonal line means:

$$TPR = FPR$$

This is like random guessing.

How to Choose a Threshold from ROC

Different applications prefer different thresholds.

Case 1: Avoid False Positives

Choose a threshold with very low FPR.

Example:

$$(FPR, TPR) = (0, 0.6)$$

No false positives, but only 60% positives found.

Case 2: Find All Positives

Choose a threshold with very high TPR.

This may allow some false positives.

What is AUC?

AUC means:

Area Under the ROC Curve

Meaning

AUC summarizes the whole ROC curve into one number.

$$0 \leq AUC \leq 1$$

AUC	Interpretation
1.0	Perfect classifier
0.5	Random guessing
Below 0.5	Worse than random

Why AUC is Useful

If we compare two models, we can compare their ROC curves.
But if there are many models, comparing curves is difficult.

AUC Advantage

AUC gives one summary score for each model.

Example:

$$AUC_{\text{Logistic Regression}} = 0.90$$

$$AUC_{\text{Naive Bayes}} = 0.85$$

Since:

$$0.90 > 0.85$$

Logistic Regression performs better overall

ROC vs Confusion Matrix

Topic	Confusion Matrix	ROC Curve
Uses	One threshold	Many thresholds
Output	TP, FP, FN, TN	TPR-FPR tradeoff
Good for	Fixed decision rule	Threshold selection
Summary score	No single curve score	AUC

Key Idea

A confusion matrix evaluates one threshold. ROC evaluates many thresholds.

Summary

- ROC is used for binary classifiers that output probabilities.
- A threshold converts probability into class label.
- Every threshold gives a confusion matrix.
- From each confusion matrix:

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

- Each threshold gives one ROC point:

$$(FPR, TPR)$$

- AUC summarizes the ROC curve.

ROC = Threshold-wise TPR vs FPR